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ORIGINAL PAPER

Shotgun metagenomics and computational profiling of the plastisphere microbiome: unveiling the potential of enzymatic production and plastic degradation

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Abstract

Plastic pollution is one of the most resilient types of pollution and is considered a global environmental threat, particularly in the marine environment. This study aimed to identify plastic-degrading bacteria from the plastisphere and their pharmaceutical and therapeutic potential. We collected samples from soil and aquatic plastisphere to identify the bacterial communities using shotgun metagenomic sequencing and bioinformatic tools. Results showed that the microbiome comprised 93% bacteria, 0.29% archaea, and 3.87% unidentified microbes. Of these 93% of bacteria, 54% were *Proteobacteria*, 23.9% were *Firmicutes*, 13% were *Actinobacteria*, and 2.1% were other phyla. We found that the plastisphere microbiome was involved in degrading synthetic and polyhydroxy alkanoate (PHA) plastic, biosurfactant production, and can thrive under high temperatures. However, no association existed between thermophiles, synthetic plastic or PHA degraders, and biosurfactant-producing bacterial species except for *Pseudomonas*. Other plastisphere inhabiting plastic degrading microbes include *Streptomyces, Bacillus, Achromobacter, Azospirillum, Bacillus, Brevundimonas, Clostridium, Paenibacillus, Rhodococcus, Serratia, Staphylococcus, Thermobifida,* and *Thermomonospora.* However, the plastisphere microbiome showed potential for producing secondary metabolites that were found to act as anticancer, antitumor, anti-inflammatory, antimicrobial, and enzyme stabilizers. These results revealed that the plastisphere microbiome upholds clinical and environmental significance as it can open future portals in a multi-directional way.

Keywords Microbiome · Next generation sequencing · Metagenomics · In-silico screening · Secondary metabolites · Environmental pollution

Introduction

A significant global ecological and environmental challenge is the pervasive pollution generated by non-biodegradable plastics (Zrimec et al. 2021). These plastics result from various product packaging and play a substantial role in environmental degradation. Accumulation of polyethylene remains in the environment can lead to the blockage of waterways, resulting in distressed water supplies, and can cause the death of several living organisms due to their consumption. It clogs drainage systems and rivers and is a leading cause of animal deaths consumed by animals looking

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for feed. Almost 0.2 million plastic garbage in Pakistan is dumped into the Indus, the most plastic-polluted river worldwide. Remorsefully, Karachi metropolis abundantly generates a maximum percentage of plastic waste, which leads to urban floods in many areas during the rainy season (Javed et al. 2022). Globally, 90% of plastic pollution is found in oceans and water bodies (Carr et al. 2022), resulting in marine microplastic pollution, a developing challenge for ecotoxicology that must be addressed (Zhai et al. 2023). Over the past 10 years, a startling 4.8–12.7 million metric tons (MMT) of plastic have made an annual journey into the ocean's depths, casting a shadow that may reach 250 MMT at the end of 2025 (Carr et al. 2022).

Recently, a novel bacterial ecosystem has been discovered that has evolved to colonize and degrade plastic soils and oceans, termed a "Plastisphere microbiome" (Javed et al. 2022). Marine plastisphere microbiome abundantly includes several species of *Pseudomonas, Bacillus,* and *Arthrobacter* that are also reported for degrading the plastic components being a part of their plastisphere habitat by secreting novel enzymes (e.g., cutinase, esterase, lipase, hydrolase, etc.) and biopolymers (e.g., biosurfactants) (Zhai et al. 2023). Moreover, marine plastisphere microbiome has several advantages over terrestrial plastisphere microbiomes in terms of plastic degradation due to diverse and specialized microbial communities adapted to the marine environment (Zrimec et al. 2021). These communities may contain thermophilic, biosurfactant-producing bacteria and other microorganisms specifically adapted to break down plastics in brine water. However, despite significant progress in identifying plastic-degrading bacteria, the taxonomic and functional diversity of the plastisphere microbiome still needs to be better studied. The use of shotgun metagenome sequencing could help to fill this knowledge gap and predict potential genes and the metabolism of microbial communities, leading to the identification of new plastic-degrading bacterial species. For example, *Ideonella sakaiensis, Bacillus thermoamylovorans*, and *Pseudomonas aeruginosa* are all thermophilic bacteria that have been shown to degrade plastic and produce biosurfactants. The ability of marine microbiomes to secrete new bioactive secondary metabolites has been demonstrated in earlier investigations. The plastic is eventually degraded by its own microbiome because the plastisphere bacteria uses the plastic components as its food supply to carry out cellular functions and reproduction. The potential contribution of a marine plastisphere microbiome to the production of secondary metabolites remains an open area for investigation. In this study, we computationally investigated the marine microbiome for the production of natural bioactive substances (Wei et al. 2023), (Plastic-Degrading Potential across the Global Microbiome Correlates with Recent Pollution Trends). Terpenes, medicinal peptides, alkaloids, lactones, and steroids are among the natural compounds. These microbial natural compounds have been documented to be versatile in their actions, such as pest resistance, antifungal, antimicrobial, antiviral, anticancer, and anti-inflammatory properties. Over 23,000 bioactive secondary metabolites are thought to be produced by marine bacteria, and they will continue to be a rich source of bioactive natural products for the ensuing century.

The discovery of marine prokaryotic secondary metabolites has drawn increasing attention in recent years, and it has been discovered that these compounds exhibit a higher chemical diversity than secondary metabolites derived from terrestrial microorganisms. To find specific marine microbial secondary metabolites, literature suggested a number of methods, including the use of genome mining techniques that also include metagenome sequencing (Wei et al. 2023). Hence, previous studies do not provide thorough guidelines and open several gaps to investigate and discover a link between plastisphere microbiome to degrade plastic and showing potential for secondary metabolites secretion. Therefore, the current study aimed to investigate the potential of marine plastisphere microbiomes in aiding plastic degradation, mainly through diverse and specialized microbial communities adapted to the marine environment, including thermophilic and biosurfactant-producing bacteria. The study also emphasized the importance of shotgun metagenome sequencing and bioinformatic tools in identifying new plastic-degrading bacterial species, understanding the taxonomic and functional diversity of the plastisphere microbiome, and predicting plastisphere microbial genes responsible for plastic degradation. Ultimately, the future aspect of the current study was to develop more efficient and sustainable plastic waste remediation strategies.

Materials and methods

Sample collection and extraction of environmental DNA (eDNA)

Samples containing polyethylene remains, including seawater, clay, plastic industry effluents, and the effluents contaminated soil, were collected within sterile falcon tubes, aseptically, from a waste dumping site at Clifton Beach, Karachi, Pakistan. As per the manufacturer's protocol, the combined representative environmental sample was subjected to genomic DNA isolation using DNeasy PowerSoil Pro Kit (Qiagen, USA). The quality of extracted DNA was analyzed using lightcycler qPCR and 1% agarose gel electrophoresis while quantified through Quant-IT PicoGreen (Invitrogen).

Library preparation and metagenome sequencing

High-quality sequencing libraries were prepared according to the manufacturer's instructions of TruSeq DNA Nano Library Prep Kit (Illumina, Inc., San Diego, CA, USA). The sequencing library was prepared by random fragmentation of the DNA or cDNA sample, followed by 5' and 3' adapter ligation. Following the fragmentation of 100 ng of genomic DNA through adaptive focused acoustic technology (AFA; Covaris), the fragmented DNA was end-repaired to create 5′-phosphorylated, blunt-ended double-stranded DNA (dsDNA) molecules. The DNA size was selected with a bead-based method, adding a single 'A' base and ligating the Truseq indexing adapters. The purified libraries were quantified using qPCR according to the qPCR Quantification Protocol Guide (KAPA Library Quantification kits for Illumina Sequencing platforms) and qualified using the Agilent Technologies 2200 TapeStation. (Agilent Technologies). The Illumina Novaseq platform was utilized to perform Whole-Genome Shotgun (WGS) sequencing of TruSeq paired-end libraries to generate 2 × 150 bp paired-end reads. This sequencing was carried out with the Illumina NovaSeq 6000 S2 Reagent Kit (Illumina, San Diego, USA), generating 100 base pair reads through 2 × 150 cycles (George 2021; Skariyachan et al. 2018).

Processing the raw sequencing data

Paired-end read Fastq files were acquired through Shotgun Metagenome Sequencing. Subsequently, Illumina sequence adapters were eliminated, and low-quality bases (< 100 bp) were removed for quality control using Trimmomatic v0.38. Quality and the impact of trimming were assessed via FASTQC v0.11.6. The DNA sequences of trimmed reads were merged and assembled through MEGAHIT v1.2.9, where a minimum K-mer length was set as 83 and a maximum K-mer length was set as 123. To determine where the contigs come from and assess the quality of the assemblies, the trimmed reads of each sample were mapped against the assemblies using bowtie2 v2.2.3. Afterward, the Genome Analysis Toolkit (GATK) v3.4.0 DepthOfCoverage module was used to compute mapping depth stepwise and insert size computation through the Picard CollectInsertSizeMetrics module. The sample contig with an average mapping depth of a value equal to or larger than 3 was stored as a separate FASTA file (Drengenes et al. 2021; Oh et al. 2018; Rölleke et al. 1996; Skariyachan et al. 2018).

Taxonomy analysis and classification

Sample contigs were assigned taxonomy using Centrifuge v1.0.4 with primary assignments set as 1, with an expected value (e) cutoff of 10^{-5}, a minimum alignment length of 50 amino acids, and a minimum percent identity of 60%. The count and portion of the contigs assigned to each taxon number were calculated at each taxon rank (kingdom, phylum, class, order, family, genus, species, and Subspecies) using in-house scripts and normalized abundance of taxon at the level of subspecies, species, and genus was calculated using the following equation. After calculating relative abundance, the ratio of abundance, compared with the total abundance sum, was calculated (Suhasa 2018).

$$\text{Abundance} = \text{Lc} * \text{D}/\text{Rt} * \text{Rl} * \text{G}$$

Lc: length of the contig; D: average mapping depth of the contig; Rt: total read count; Rl: average read length; G: average genome size of the taxon.

Some contigs were not allotted a respective taxon by the Centrifuge v1.0.4 because they were not classified on the previously existing biological databases. These contigs were binned to classify through CONCOCT v1.1.0 by targeting characteristics of the sequences, even without a length threshold (Hamdani 2019).

Gene prediction and annotation

Besides other bioinformatics analyses, the most significant analysis of this study was the prediction of bacterial genes involved in plastic degradation. The contigs were subjected to Prokka v1.13 for genes prediction that were annotated using EggNOG-mapper v2.0.1 based on EggNOG database v5.0. The EggNOG database is a hierarchical, functionally, and phylogenetically annotated orthology resource. Like the COG database, genes are categorized into 25 categories (Alneberg et al. 2014; Lenfant et al. 2013; Seemann 2014).

Enrichment of alpha and beta hydrolases

Alignment of predicted coding sequences (pCDS) from the metagenome sample was carried out against the ESTHER database of alpha/beta hydrolases (depolymerases, esterases, lipases, and cutinases) that were also reported for plastic degradation previously (DiGiacomo et al. 2022). CD-HIT version 4.7 was used to cluster protein sequences with default settings to reduce redundancy within the database. The modified database used included 10 protein families containing a total of 17,312 protein sequences. Alignments were performed using the command-line version of BLASTp, as implemented in BLAST$^+$ version 2.6.0, with an expected value (e) cutoff of 10^{-5}, a minimum alignment length of 50 amino acids, and a minimum percent identity of 60%.

Association of plastisphere microbiome with significant potential bacterial activities

Using the available data on ThermoBase (DiGiacomo et al. 2022; Oliveira et al. 2015), the taxonomic profile of shotgun metagenome was analyzed for the plastisphere microbiome to have the potential for striving in high temperatures. Similarly, the plastisphere microbiome was further analyzed for the potential of degrading synthetic and bioplastics, and biosurfactant production, with the help of BioSurfDB (Oliveira et al. 2015), and PlasticDB (Gambarini et al. 2022). Using AntiSmash database (Blin et al. 2021), secondary metabolites of the plastisphere microbiome were predicted.

Data visualization

Heatmaps and circus-plot were constructed using R package pHeatmap (Kolde 2019), and Circlize (Brors 2014), on RStudio v 4.3.3.2 (Team 2021).

Data availability

The metagenome profile is submitted on NCBI, with the BioSample ID 410658 and BioProject ID PRJNA764827.

Results

Metagenomics sequencing and assembly

A total of 38,078,676 reads were generated by high throughput shotgun metagenome sequencing. Average contig lengths of about 632,427 bp to 200 bp, and approximately 99.52% high-quality read Q20 and Q30 were generated. The high-quality read was assembled in 17,312 contigs, an N50 score of 2263, and the minimum length of contigs was 200 bp.

Estimation of microbial community structure

Metagenomic data revealed the presence of 875 distinct bacterial genera in the collected environmental sample, including 3.87% of unidentified microbes, 0.29% of archaea, and 93% of bacterial diversity. The bacterial community included 54% *Proteobacteria*, 23.9% *Firmicutes*, 13% *Actinobacteria*, and 2.1% other bacteria such as *Aquificae* (0.005%), *Bacterioidetes* (0.71%), *Balneolaeota* (0.075%), *Chlamydiae* (0.01%), *Chlorobi* (0.005%), *Chloroflexi* (0.017%), *Cyanobacteria* (0.075%), *Deinococcus* (0.063%), *Euryarchae* (0.288%), *Fibrobacteres* (0.011%), *Fusobacteria* (0.040%), *Kiritimatiellaeota* (0.005%), *Lentisphaerae* (0.005%), *Nitrospinae* (0.005%), *Planctomycetes* (0.092%), *Rhodothermaeota* (0.005%), *Spirochaetes* (0.035%), Synergistetes (0.006%), *Tenericutes* (0.063%), *Thaumarchaeota* (0.006%), *Thermotogae* (0.011%), and *Verrucomicrobia* (0.017%), as shown in Fig. 1.

Association of plastisphere microbiome with significant potential bacterial activities

Only *Pseudomonas* showed the capability to thrive under high temperatures, degradation of PHA, synthetic plastic and biosurfactant production. Like *Pseudomonas, Streptomyces* exhibited thermophilic characteristics and was also involved in the degradation of PHA. Besides, *Bacillus, Mycobacterium, Rugeria, Streptomyces,* were found to be biosurfactant producers along with having a thermophilic nature. However, *Clostridium, Mycobacterium, Enterococcus, Nocardioides, Pseudomonas, Staphylococcus,* and *Rhodococcus* could produce biosurfactants and degrading synthetic plastics. The association of plastisphere microbiome with certain potentials is summarized in Fig. 2.

Overall functional annotation

The predicted genes were translated into respective proteins using the eggnog mapper tool. A total of 2069 proteins were mapped, and 195 proteins did not match with previously identified proteins. As shown in Fig. 3, 1973 proteins were found in the pre-existing protein database, which included 534 proteins with unknown functions, facilitating the idea for conducting extensive research studies to identify their functions. On the other hand, the metagenome contains many proteins involved in transcription, translation, biogenesis, inorganic ion transport and metabolism, amino acid transport and metabolism, carbohydrate transport, and metabolism, energy production and conversion, and other secondary metabolites. Secondary metabolites include the enzymes encompassing plastic degradation and other important clinical and industrial biotechnology-related activities.

Secondary metabolites

The production of secondary metabolites from the plastisphere microbiome was predicted using in-silico analysis. The predicted results of secondary metabolites and their applications are summarized in the Circos plot (Fig. 4).

The predicted results reveal that plastisphere microbiome potentially could produce ectoine, lanthipeptides, cyclic lactone autoinducers, transATPKS and NRPS, ranthipeptides, terpenes, and polyketide synthase.

Enrichment of alpha/beta hydrolases

Hydrolases, such as cutinase, depolymerase, esterase, and lipase encoding genes were abundantly found in the plastisphere microbiome. Figure 5 shows alpha and beta

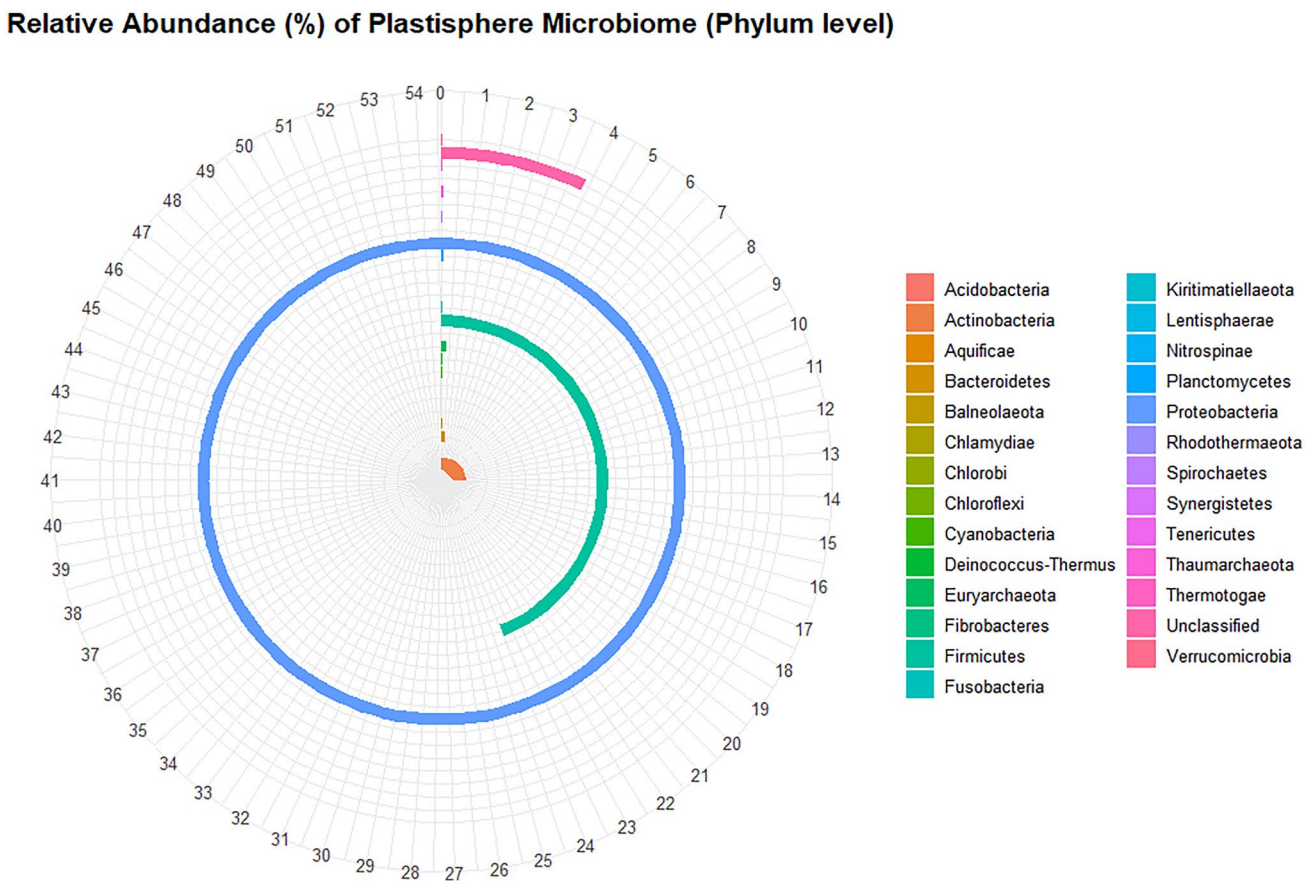


Fig. 1 Composition of the plastisphere microbiome: relative abundance (%) of microbial taxa at Phylum level within a specific microbiome. Each colored segment represents a distinctive phylum, and its size corresponds to the percentage of the microbiome it constitutes. Higher segments indicate a greater abundance of that target phylum within the microbiome

hydrolases secretion with respective percentages, found in the plastisphere microbiome.

Gene prediction

The diverse variety of genes was predicted in the plastisphere microbiome using Prokka. Besides the genes having essential cellular metabolic functions, a gene set was found that involved the plastic degradation phenomenon. These genes encode various hydrolases and oxidases, as shown in Table 1.

Discussion

The microbial colonization of plastics established in the marine environment has been well-documented (Wallbank et al. 2022). Still, there is a need for a more thorough study identifying the plastic-degrading microbial communities and their contribution to some prominent attributes encompassing in-silico biosynthesis of industrially and clinically essential metabolites. This research is the first-ever study of the plastisphere microbiome in Karachi, Pakistan, making it particularly significant. The plastisphere microbiome was investigated in various polyethylene (PE) debris-contained environments, including marine, seashore, and plastic dumping sites. *Ideonella sakaiensis,* and the species of *Marinobacter, Halomonas, Pseudomonas,* and *Bacillus* were abundant in the plastisphere microbiome. This established an understanding of microbial communities of the plastisphere. We have also predicted that the analyzed plastisphere microbiome was involved in degrading plastic and could thrive in high-temperature environments, besides exhibiting the potential to synthesize industrially and clinically important secondary metabolites.

From *Pseudomonas, Enterococcus, Mycobacterium, Clostridium, Rhodococcus, Staphylococcus and Streptomyces* only *Pseudomonas* and *Streptomyces* showed degradation capability for polyhydroxyalkanoates (PHA). *Pseudomonas* were also predicted to degrade synthetic plastic. Moreover, *Pseudomonas* and *Streptomyces* showed biosurfactant production, ability to thrive in high temperatures,

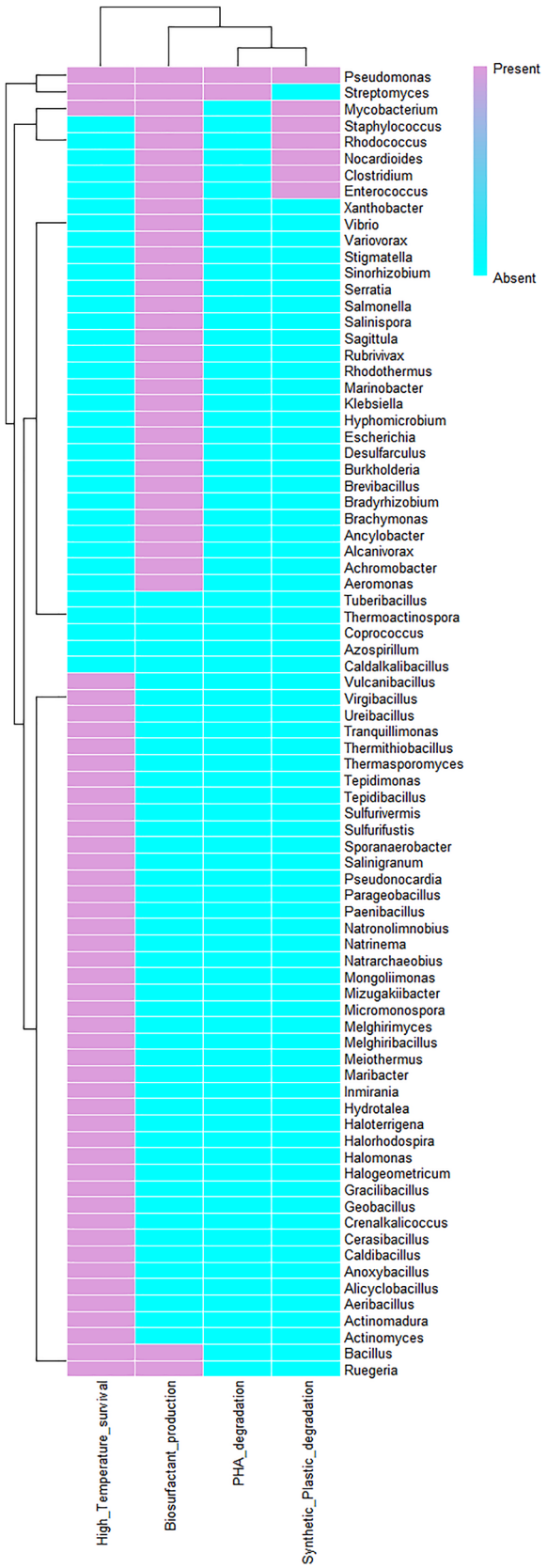


Fig. 2 Heatmap depicting the association of the plastisphere microbiome with important characteristics, such as thermophilic nature, biosurfactant production, and degradation of PHA and synthetic plastics

and PHA degradation. It has already been reported that thermophilic *Pseudomonas* species are involved in biosynthesis and biodegradation of PHA (Mozejko-Ciesielska et al. 2019). However, the degradation of PHA by microorganisms is part of the natural carbon cycle. It allows for the release of carbon and energy stored in the polymer back into the ecosystem, enabling other organisms to utilize these resources for growth and metabolism, giving off other metabolites and enzymes for cellular activities (Volova et al. 2010). Unlike chemical surfactants, biosurfactants are a kind of secondary metabolites that provide a favorable environment for the microbes to carry out their significant activities, for instance, bioremediation of plastics (PHA, PE, PET, and so on) (Bhadra et al. 2022). According to the literature, *Bacillus subtilis* and *Pseudomonas aeruginosa* degrade low-density polyethylene (LDPE) with the help of their biosurfactant-producing capability (Nnaji et al. 2021). From the plastisphere microbiota, *Pseudomonas, Staphylococcus, Rhodococcus, Nocardioides, Clostridium,* and *Enterococcus* were predicted for biosurfactant production and synthetic plastic degradation. However, around 50% of biosurfactants-producing bacteria showed an association with plastic-degrading bacterial species. Moreover, *Cyanobacteria, Bacterioidetes, Proteobacteria, Acidobacteria*, and *Actinobacteria* have already been reported to be a part of the plastisphere microbiome. This study also concluded that the indicator species of a plastisphere were predominated by *Proteobacteria, Actinobacteria, Myxococcota, Chloroflexi,* and *Firmicutes* (Sun et al. 2022). Of which, *Rhodococcus, Staphylococcus, Arthrobacter, Bacillus, Pseudomonas, Brevibacillus,* and *Streptomyces* have so far been reported for degrading polyethylene terephthalate and other plastics (Benavides Fernández et al. 2022; Ru et al. 2020).

Numerous thermophilic microbes have shown high potential for plastic degradation due to their ability to grow at high temperatures and produce numerous oxidase and hydrolase enzymes (Dang et al. 2018). Previous studies showed the degradation of polyethylene (PE), polystyrene foam (PS) and polyethylene terephthalate (PET) by thermophilic microbial consortium, composed of *Penicillium raperi, Aspergillus flavus*, *Penicillium glaucoroseum* and *Pseudomonas* spp. in the presence of biosurfactant (rhamnolipid) (Taghavi et al. 2022). Biological surfactants possess several potential advantages over their chemically manufactured counterparts, including lower toxicity, biodegradability and effectiveness at extreme temperatures, pH and salinity (Midhun et al. 2015). In the investigated plastisphere microbiome, there were 46 bacterial genera that showed involvement in striving and thriving under high temperatures (Fig. 2). Of which, *Pseudomonas* and *Streptomyces* were found to be PHA degraders; and *Mycobacterium* and *Pseudomonas* were synthetic plastic degraders. Moreover, *Bacillus, Mycobacterium, Pseudomonas, Rugeria,* and *Streptococcus* were found

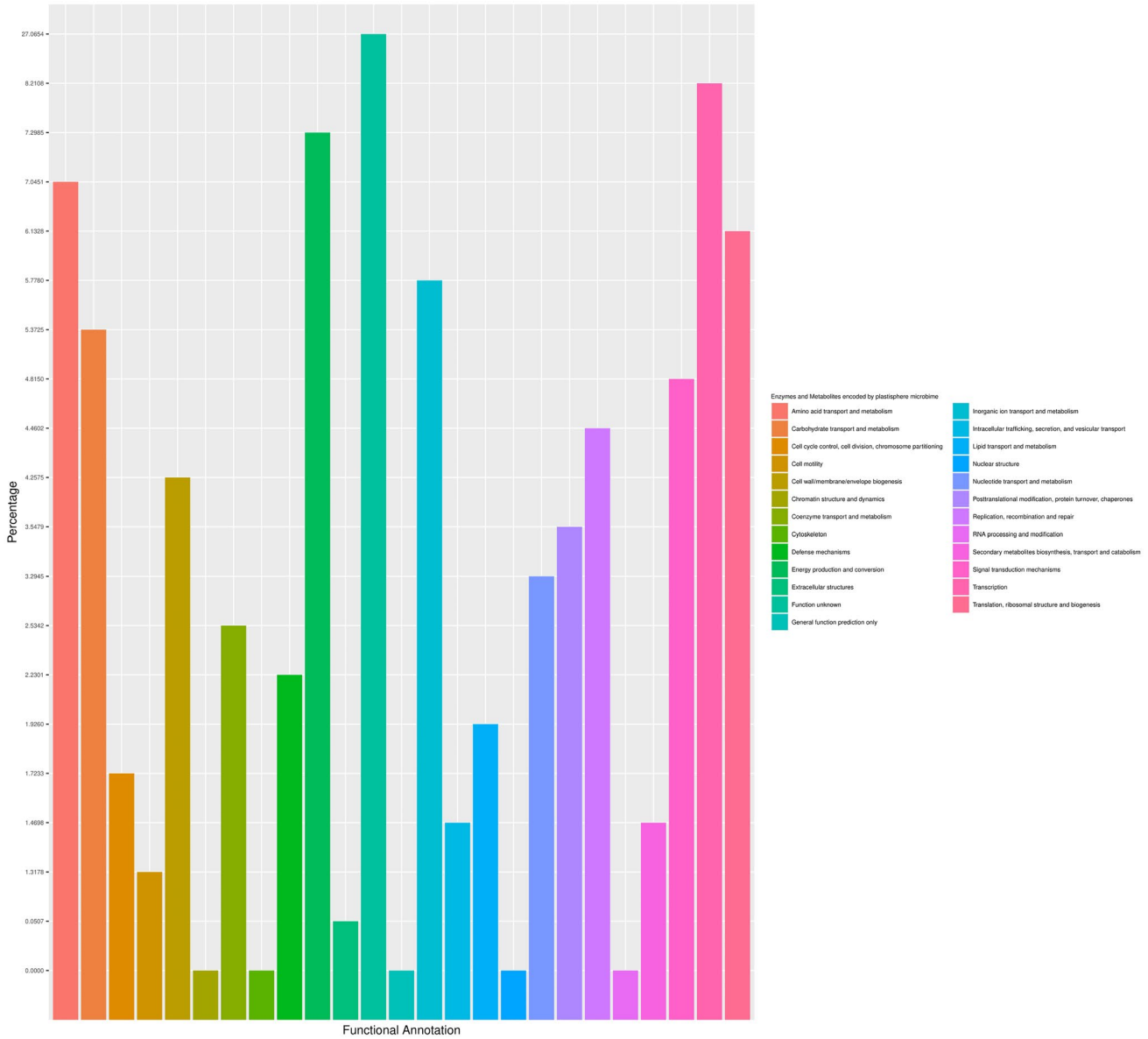


Fig. 3 Functional annotations (abundance %) of the plastisphere microbiome. Each bar represents a functional category, and the length corresponds to the abundance or prevalence of that category in the microbiome

to be biosurfactant producers besides being thermophilic in nature. The results of our study are also in accordance with Pang et al. (2023) who reported that plastic debris could be a promising source of plastic-degrading species. Furthermore, the analyzed plastisphere microbiome was found to secrete enzymes involved in plastic degradation, including peroxidase (42%), lipase (19%), esterase (15%), depolymerase (12%), and cutinase (17%). These enzymes are known for breaking down plastics, which likely contributes to plastic degradation by the microbiome, as reported in "The plastisphere microbiome in alpine soils alters the microbial genetic potential for plastic degradation and biogeochemical cycling".

Literature also confirms esterase production by *Thermobifida*, *Bacillus*, *Nocardia*, *Saccharomyces*, and *Clostridium*. Cutinase production has been observed in species like *Halotolerans* and *Thermobifida* (Khairul Anuar et al. 2022). Lipase production is attributed to *Burkholderia*, *Pseudomonas*, and *Bacillus*. Additionally, species of *Pseudomonas*, *Actinobacteria*, *Burkholderia*, *Zymomonas*, *E. coli*, and *Salmonella* are known to produce peroxidase (Khairul Anuar et al. 2022). Interestingly, all the above-mentioned microbes were identified in the plastisphere microbiome. This suggests that these microbes could play a potential role in the degradation of plastics. However, we found variations in the results when we predicted the genes

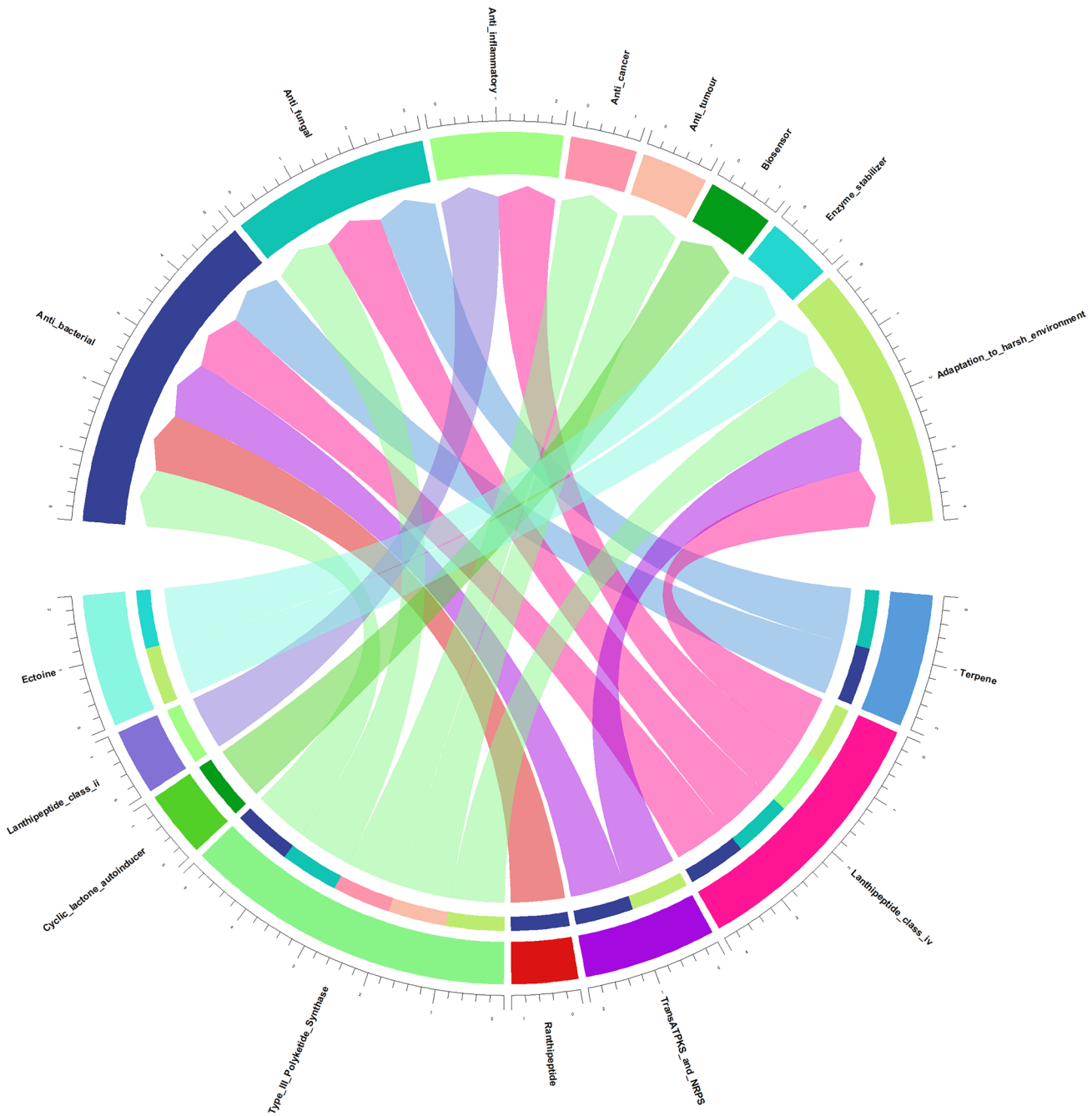


Fig. 4 Circos plot illustrating predictions for the secretion of secondary metabolites (bioactive compounds) by the plastisphere microbiome. Each block in the Circos plot represents a predicted secondary metabolite within the plastisphere microbiome, delineating its respective significant function through connecting chords

involved in the degradation of plastics. Besides the genes responsible for encoding esterase (EstA), cutinase (cut-1, cut-2), lipase (LIP1, LIP2), and peroxidase (lignin peroxidase: dypA), other genes included mco that is reported for encoding multicopper oxidases. Moreover, est and pet were also found in the microbiome, which are responsible for secreting carboxyl esterase and polyethylene terephthalate depolymerase, respectively. All these enzymes have been reported for degrading plastics (Zhang et al. 2023). This shotgun microbiome also showed potential for alpha and beta hydrolase enzyme secretion, such as peroxidases, lipases, esterases, depolymerases, and cutinases. Cutinases, lipases, and esterases are reported for degrading Polyethylene terephthalate (PET), polylactic acid, and polyurethane,

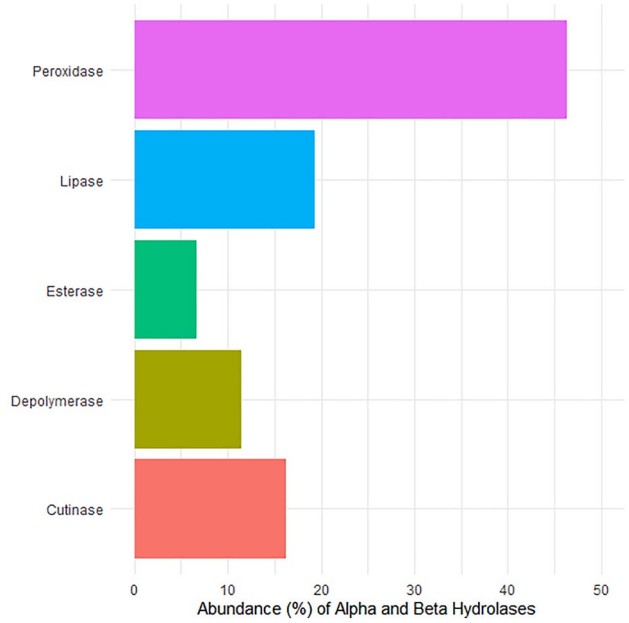


Fig. 5 Bar chart illustrating the predicted abundance (%) of Alpha and Beta hydrolases within the plastisphere microbiome

respectively (Srinivasan et al. 2021; Yamashita et al. 2005), (Akash Karthikeyan et al. 2022a, b). However, polyethylene and polypropylene are degraded by peroxidase enzymes. Depolymerases are the family of enzymes reported to degrade all forms of plastic (Kaushal et al. 2021; Mohanan et al. 2020). The role of depolymerase enzymes in plastic degradation has been extensively studied, and researchers have identified various depolymerases with specific plastic-degrading capabilities (Wallbank et al. 2022).

Secondary metabolites are beneficial for human health (Ali et al. 2023). Besides primary metabolites and enzymes, the plastisphere microbiome showed potential to produce secondary metabolites, including terpene, lanthipeptide class IV, transATPKS and NRPS, ranthipeptide type III, polyketide synthase, cyclic lactone autoinducer, lanthipeptide class II, and ectoine. Of these, lanthipeptide class IV, transATPKS and NRP, and ectoine were the primary sources to stabilize the bacterial enzymes to thrive in harsh conditions, such as extremely high or low temperatures, salt or acidic concentrations, and osmotic pressures. Several microbes from marine environments have previously been discovered for their novel bioactivities (natural products) production (Karthikeyan et al. 2022a, b). Overall, the plastisphere microbiome was predicted to produce secondary metabolites with significant features, including antibacterial, antifungal, anti-inflammatory, anticancer, antitumor, and enzyme stability effects, that can be widely utilized to cover the domain of red biotechnology. Ectoin, a molecule synthesized by the annotated microbiome (Fig. 4), is predicted to have a protective role in harsh environments. It ensures enzyme stability and activity, making it a valuable macromolecule shield (Yoshida et al. 2016; Zhang et al. 2022).

The plastisphere microbiome was also predicted to degrade PHA. Still, there was no synergistic association between the degradation of synthetic plastic, PHA, biosurfactant production, and the capabilities of bacteria to strive and thrive at high temperatures. Several microbes belonging to phyla Proteobacteria and Bacteroidetes are reported for the secretion of secondary metabolites, including terpenes, alkaloids, and ectoine (Di Costanzo et al. 2021).

In the intriguing world of secondary metabolites, one remarkable hallmark of the microbiome is its potential for secreting ectoine. Compatible solutes, like ectoins or di-myo-inositol phosphate, help cells adapt to osmotic changes and protect against freezing, UV radiation, high temperature, and macromolecule damage. Phosphate was common in mesophilic bacteria, while negatively charged solutes like di-myo-inositol phosphate and mannosyl glycerate are found in hyper/thermophilic bacteria and archaea.

Table 1 Predicted genes involved in plastic degradation

Name	Description	Predicted gene
Putative HAD hydrolase YfnB	Hydrolase	yjjG
Polyphenol oxidase	Multicopper oxidase	yfiH
Hypothetical protein	Multicopper oxidase	
Multicopper oxidase mco	Multicopper oxidase	Mco
Lignin peroxidase	Peroxidase	dypA
Esterase	Alpha/beta hydrolase	EstA
PETase (Polyethylene terephthalatease)	Depolymerase	pet
Carboxylesterase	Alpha/beta hydrolase family	Est
Lipase	Alpha//beta hydrolase	LIP1 and LIP2
Protease	Hydrolase	apr
Cutinase		cut-1 and cut-2

Bacterial ectoin synthesis increases under hypo- or hyperosmotic stress (Di Costanzo et al. 2021; Kauth and Trusova 2022).

Moving to the lanthipeptide secretion, these fascinating compounds exhibit a broad spectrum of antimicrobial activities, encompassing antibacterial and antifungal functions and antiviral, antinociceptive, antiallodynic, and morphogenetic properties. Lanthipeptides identified so far are made mainly by Gram-positive bacteria, such as pinensins, which are fungicidal and found in bactericides (Caetano et al. 2020; Castro et al. 2021). Terpenes, polyketides, nonribosomal peptides, and RiPPs (ribosomally produced and post-translationally modified peptides) are the most common bioactive secondary metabolites in bacteria that include the class Halobacteria, also known as haloarchaea. Because they are halophiles, haloarchaea can only exist in a small range of ambient salt concentrations. They are the dominant prokaryotic species in hypersaline, thalassohaline, and athalassohaline habitats (such as marine settings). They have also been thought to make ideal candidates for life on Mars since they can flourish in stromatolites and ancient halite (Najjari et al. 2015; Oren 2014; Yu et al. 2013).

Interestingly, natural terpenes such as linalool (L), geraniol (G), and geranyl acetate (GA) have been reported for synthesizing poly(3-hydroxybutyrate), which is a kind of bioplastic (Mangeon et al. 2018; Turco et al. 2020). Actinobacteria are a treasure trove of lanthipeptides, as revealed by genome sequencing, which has shown an abundance of lanthipeptide biosynthetic gene clusters. The hexapeptide isolated from *Streptomyces rochei* Sal35, the roseocin isolated from *Streptomyces roseosporus*, the mathermycin isolated from *Marinactinospora thermotolerans,* the cebulantin isolated from *Saccharopolyspora cebuensis*, the microbisporicin isolated from *Microbispora corallina*, and others has all been reported in previous studies (Chen et al. 2017; Fernández-Martínez et al. 2015; Gomes et al. 2017; Li et al. 2021; Moon et al. 2019; Singh et al. 2020; Uritskiy et al. 2019; Vikeli et al. 2020; Walker et al. 2020; Xu et al. 2020).

Conclusion

The marine plastisphere microbiome, sourced from diverse environments encompassing brine water with microplastics, plastic waste disposal sites, and clay deposits at coastal zones, represents a repository of significant bioactive compounds. In addition to its pivotal role in mitigating plastic pollution through bioremediation, this microbiome exhibits substantial potential for biosurfactant production, particularly rhamnolipids, which have demonstrated remarkable efficacy in plastic degradation. Furthermore, this plastisphere microbiome showcases a notable capacity for the biosynthesis of secondary metabolites, a critical facet in advancing pharmaceutical and enzyme systems, offering solutions that streamline clinical and industrial processes. This study firmly establishes a robust connection between the marine plastisphere microbiome and its ability to produce industrially and clinically relevant bioactive compounds, all while effectively addressing the pervasive issue of plastic pollution. Notably, *Pseudomonas* emerges as a critical player among these microorganisms' demonstrating involvement in plastic degradation, biosurfactant production, and exhibiting thermophilic characteristics. Consequently, this study pioneers a forward-looking approach towards comprehensively investigating the functions and attributes of these secondary metabolites to harness their potential in a targeted and constructive manner. More research should be conducted to further validate these results.

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Declarations

Competing interests The authors declared that they have no competing interests.

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